Main points :

Strum - Liouville

$$\left[\frac{d}{dx}\left(p(x)\frac{d\phi}{dx}\right) + q(x)\,\phi(x) + \lambda\sigma(x)\,\phi(x) = 0 \right]$$

+ some B.C

$$\begin{cases} \phi = 0 & \text{Dirichlet} \\ \dfrac{d\phi}{dx} = 0 & \text{Neumann} \\ \dfrac{d\phi}{dx} = \pm h\,\phi & \text{Robin} \end{cases} \Bigg\} \text{ homogens}$$

$$\begin{cases} \left.\begin{aligned} \phi(-L) &= \phi(L) \\ \phi'(-L) &= \phi'(L) \end{aligned}\right\} \text{periodic} \\ |\phi(0)| < \infty \end{cases}$$

R.S.L $\quad \beta_1 \phi(a) + \beta_2 \phi'(a) = 0$

$\beta_3 \phi(b) + \beta_4 \phi'(b) = 0$

- $\lambda \in \mathbb{R}$
- $\exists$ infinite eigenvalues

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n \leq \cdots$$

- to each eigenvalue $\exists$ a unique eigenfunction $\phi_n(x)$. it has exactly $(n-1)$ zeros in the interval $a < x < b$.
- ϕ_n are complete

$$f(x) \sim \sum a_n \phi_n(x)$$

$f(x)$ is piecewise continuous in the interval.

- Eigenfunctions are orthogonal.

$$\int_a^b \phi_n(x)\, \phi_m(x)\, \sigma(x)\, dx = 0$$

if $\lambda_n \neq \lambda_m$.

- orthogonality in singular cases (if $p(x) = 0$ at the boundaries and P.B.C.
- uniqueness of eigenfunctions in P.B.C.

Generalized fourier series:

$$f(x) = \sum_{n=0}^{\infty} c_n \phi_n(x)$$

$$\langle \phi_m | f(x) \rangle = \sum_{n=0}^{\infty} c_n \underbrace{\langle \phi_n | \phi_m \rangle}_{\|\phi_m\|^2 \delta_{nm}}$$

$$\Rightarrow \quad c_m = \frac{\langle \phi_m | f(x) \rangle}{\|\phi_m\|^2}$$

$$= \frac{\int_a^b \sigma(x) \phi_m(x) f(x)\, dx}{\int_a^b \sigma(x) \phi_m^2(x)}$$

} can be different for different B.C.

Example: Generalized Bessel Series.

$$f(x) = \sum_{i=1}^{\infty} C_i \, J_n(\alpha_i x)$$

- $J_n(\alpha b) = 0$

$$\|J_n\|^2 = \frac{b^2}{2} J_{n+1}^2(\alpha_n b)$$

- $h J_n(\alpha b) + \alpha b J_n'(\alpha b) = 0$

$$\|J_n(\alpha_i x)\|^2 = \frac{\alpha_i^2 b^2 - n^2 + h^2}{2\alpha_i^2} J_n^2(\alpha_i b).$$

- $J_0'(\alpha b) = 0$

$$\|J_0(\alpha_i x)\|^2 = \frac{b^2}{2} J_0^2(\alpha_i b)$$

& $\|1\|^2 = \frac{b^2}{2}$ for $\boxed{\alpha_1 = 0}$

Fourier Legendre

$$f(x) = \sum_{n=0}^{\infty} C_n P_n(x)$$

$$C_n = \frac{2n+1}{2} \int_{-1}^{1} f(x) P_n(x) \, dx$$

Heat equation

$$PDE: \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

BC & IC

problem: Given that $J_0(k_n) = 0$

Show that

$$1 - x^2 = 8 \sum_{n=1}^{\infty} \frac{J_0(k_n x)}{k_n^3 J_1(k_n)}$$

Sol. $1-x^2 = \sum_{n=1}^{\infty} c_n J_0(k_n x)$

$$c_n = \frac{2}{J_1^2(k_n)} \int_0^1 x J_0(k_n x)(1-x^2)\,dx$$

$$\int_0^1 x J_0(k_n x)\,dx$$

$u = k_n x$

$du = k_n\,dx$

$$\Rightarrow \int_0^{k_n} \frac{u}{k_n^2} J_0(u)\,du$$

$$= \frac{1}{k_n^2}\int_0^{k_n} u J_0(u)\,du$$

$$= \frac{1}{k_n^2}\int_0^{k_n b} \frac{d}{du}\left[u J_1(u)\right] du$$

$$= \frac{1}{k_n^2}\left[k_n b J_1(k_n)\right]$$

$$\int_0^1 x^3 J_0(k_n x)\, dx$$

$$\int_0^{k_n} \frac{u^3}{k_n^4} J_0(u)\, du$$

$$= \frac{1}{k_n^4} \int_0^{k_n} u^3 J_0(u)\, du$$

$$= \frac{1}{k_n^4} \left(\left((k_n)^3 - 4(k_n) \right) J_1(k_n) + 2 k_n^2 b^2 J_0(k_n) \right)$$

$$= \left(\frac{1}{k_n} - \frac{4}{k_n^3} \right) J_1(k_n)$$

$$\Rightarrow \frac{4}{k_n^3} J_1(k_n)$$

$$\Rightarrow C_n = \frac{8}{k_n^3 J_1(k_n)}$$

problem:

Given; $J_0(k_n) = 0$

show that $\ln(x) = -2 \sum_{n=1}^{\infty} \frac{J_0(k_n x)}{[k_n J_1(k_n)]^2}$

$$\ln(x) = \sum_{n=1}^{\infty} C_n J_0(k_n x).$$

$$C_n = \frac{2}{J_1^2(k_n)} \int_0^1 x \ln(x) J_0(k_n x)\, dx$$

$$\int_0^1 x \ln(x) J_0(k_n x)\, dx = \frac{1}{k_n^2} \int_0^{k_n} u \ln\left(\frac{u}{k_n}\right) J_0(u)\, du$$

$u = \ln(\frac{u}{k_n})$ $\qquad v' = u J_1(u)$

$du = \frac{du}{u}$ $\qquad dv = u J_0(u)\, du$

$$\Rightarrow \frac{u \ln\left(\frac{u}{k_n}\right) J_1(u)}{k_n^2} \Bigg|_0^{k_n} - \frac{1}{k_n^2} \int_0^{k_n} J_1(u)\, dx$$

$$\lim_{x \to 0} x \ln(x) = \lim_{x \to 0} \frac{\ln(x)}{1/x} = \frac{1/x}{-1/x^2}$$

$$= \lim_{x \to 0} -x = 0$$

$$= -\frac{1}{k_n^2} \int_0^{k_n} J_1(x)\, dx$$

$$-\frac{d}{dx}\left[J_0(x) \right] = J_1(x)$$

$$= \frac{1}{k_n^2} \left[\cancel{J_0(k_n)}^{\,0} - J_0(0) \right]$$

$$= -\frac{1}{k_n^2}$$

$$\Rightarrow \quad C_n = -\frac{2}{\left[k_n J_1(k_n) \right]^2}$$

problem: Given $J_p'(k_n) = 0$

Show that

$$x^p = 2 \sum_{n=1}^{\infty} \frac{k_n J_{p+1}(k_n)}{(k_n^2 - p^2)[J_p(k_n)]^2} J_p(k_n x)$$

Sol.

$$x^p = \sum_{n=0}^{\infty} c_n J_p(k_n x)$$

$$c_n = \frac{2k_n^2}{(k_n^2 - n^2) J_{p+1}^2(k_n)} \int_0^1 x^{p+1} J_p(k_n x)\, dx$$

$$= \frac{2k_n^2}{(k_n^2 - n^2) J_{p+1}^2(k_n)} \left[\frac{1}{k_n^{p+2}} x^{p+1} J_{p+1}(x) \right]_0^{k_n}$$

$$= \frac{2k_n J_{p+1}(k_n)}{(k_n^2 - n^2) J_{p+1}^2(k_n)}.$$

develop legendre series for

$$f(x) = \begin{cases} -1 & -1 \le x < 0 \\ 1 & 0 < x \le 1 \end{cases}$$

$$f(x) = \sum_{n=0}^{\infty} C_n P_n(x)$$

$$C_n = (n+1/2) \int_{-1}^{1} f(x) P_n(x) \, dx = 0$$

$f(x)$ is odd so $n = 0, 2, 4 \ldots$ $C_n = 0$

$n = 2k+1$ & n

$$C_n = (n+1/2) \int_{-1}^{1} f(x) P_n(x) \, dx$$

$$= (n+1/2) \left[2 \int_{0}^{1} P_n(x) \, dx \right]$$

$$P_n(x) = \frac{1}{2n+1}\left[P'_{n+1}(x) - P'_{n-1}(x) \right]$$

$$\Rightarrow \int_0^1 P_n(x)\,dx = \frac{1}{2n+1}\left[P_{n+1}(x) - P_{n-1}(x) \right]_0^1$$

Since $P_n(1) = 1$

$$= \frac{1}{2n+1}\left[P_{n+1}(0) - P_{n-1}(0) \right]$$

$n = 2k+1$

$$\Rightarrow C_{2k+1} = P_{2k}(0) - P_{2k+2}(0)$$

$$P_{2n}(0) = \binom{-1/2}{n} = \frac{(-1)^n (2n)!}{2^{2n}\,(n!)^2}$$

$$\Rightarrow C_{2k+1} = \frac{(-1)^k (2k)!}{2^{2k}(k!)^2} - \frac{(-1)^{k+1}(2k+2)!}{2^{2k+2}\,[(k+1)!]^2}$$

$$= \frac{(-1)^k\,(2k)!\,(4k+3)}{2^{2k+1}\,k!\,(k+1)!}$$

Apply to

$$f(x) = \begin{cases} 1 & -1 \le x < 0 \\ x & 0 < x \le 1 \end{cases}$$

$$C_n = (n+1/2) \int_{-1}^{1} f(x)\, P_n(x)\, dx$$

$$= (n+1/2)\left[\int_{-1}^{0} P_n(x)\, dx + \int_{0}^{1} x\, P_n(x)\, dx \right]$$

$$x\, P_n = \frac{1}{2n+1}\left[(n+1)\, P_{n+1} + n\, P_{n-1} \right]$$

$$\int_0^1 x P_n\, dx = \frac{n+1}{2n+1} \int_0^1 P_{n+1}\, dx + \frac{n}{2n+1} \int_0^1 P_{n-1}\, dx$$

$$P_n(x) = \frac{1}{(2n+1)} \left[P'_{n+1}(x) - P'_{n-1}(x) \right]$$

$$\int_a^b dx\, P_n(x) = \frac{1}{(2n+1)} \left[P_{n+1}(x) - P_{n-1}(x) \right]_a^b$$

$$\Rightarrow C_n = (n+1/2) \left[\frac{1}{2n+1} \left(P_{n+1}(x) - P_{n-1}(x) \right) \Big|_{-1}^{0} \right.$$

$$+ \frac{1}{2n+1} \left[\frac{(n+1)}{2n+3} \left(P_{n+2}(x) - P_n(x) \right) \Big|_0^1 \right.$$

$$\left. \left. + \frac{n}{2n-1} \left(P_n(x) - P_{n-2}(x) \right) \Big|_0^1 \right] \right]$$

Method of seperation of variables

$$\text{PDE:} \quad \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

$$\text{B.C} : \quad \left.\begin{aligned} u(0,t) &= 0 \\ u(L,t) &= 0 \end{aligned}\right\}$$

$$\text{IC} : \quad u(x,0) = f(x)$$

- Heat conduction in a Rod w/ insulated ends

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

$$\text{BC:} \quad \frac{\partial u}{\partial x}(0,t) = 0$$

$$\frac{\partial u}{\partial x}(L,+) = 0$$

$$\text{IC} : u(x,0) = f(x)$$

& find steady-state solution.

Sol: $u(x,t) = \phi(x)\, G(t)$

$$\Rightarrow \quad \phi(x) \frac{dG(t)}{dt} = k\, G(t) \frac{\partial^2 \phi(x)}{\partial x^2}$$

$$\Rightarrow \quad \frac{1}{k} \frac{1}{G(t)} \frac{dG(t)}{dt} = \frac{1}{\phi(x)} \frac{\partial^2 \phi(x)}{\partial x^2} = -\lambda$$

$$\Rightarrow \boxed{1} \quad \frac{dG(t)}{dt} = -k G(t) \lambda$$

$$\Rightarrow \boxed{G(t) = c e^{-k\lambda t}} \leftarrow \text{unbounded time interval } t > 0$$

$$\frac{1}{\phi(x)} \frac{\partial^2 \phi(x)}{\partial x^2} = -\lambda$$

$$\Rightarrow \quad \frac{\partial^2 \phi(x)}{\partial x^2} + \lambda \phi(x) = 0$$

from $\quad G(t) \frac{\partial \phi(0)}{\partial x} = G(t) \frac{\partial \phi(L)}{\partial x} = 0 \quad$ B.C

$$\Rightarrow \quad \frac{\partial \phi(0)}{\partial x} = \frac{\partial \phi(L)}{\partial x} = 0$$

Case I: $\lambda > 0$

$$\phi(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$$

$$\frac{d\phi(x)}{dx} = -C_1\sqrt{\lambda}\sin(\sqrt{\lambda}x) + C_2\sqrt{\lambda}\cos(\sqrt{\lambda}x)$$

$$\left.\frac{d\phi(x)}{dx}\right|_{x=0} = C_2\sqrt{\lambda} = 0 \qquad \lambda > 0$$

$$\Rightarrow C_2 = 0$$

$$\phi(x) = C_1 \cos(\sqrt{\lambda}x)$$

$$\left.\frac{d\phi(x)}{dx}\right|_{x=L} = -C_1\sqrt{\lambda}\sin(\sqrt{\lambda}L) = 0$$

$$\Rightarrow \sqrt{\lambda}L = n\pi \qquad n = 1, 2, 3, \ldots$$

$$\Rightarrow \lambda = \frac{n^2\pi^2}{L^2}.$$

$\lambda = 0$

$$\phi(x) = C_1 x + C_2$$

$$\frac{d\phi(x)}{dx} = C_1 = 0.$$

$$\phi(x) = \boxed{C_2}$$

$\lambda < 0$: the time dependent part grows exponentially which is unphysical.

$$\Rightarrow \quad u(x,t) = A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{L}\, e^{-(n\pi/L^2)kt}$$

$$= \boxed{\sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L^2)kt}}$$

Steady state solution:

$$\lim_{t\to\infty} u(x,t) = A_0 = \frac{1}{L}\int_0^L f(x)\,dx$$

Heat conduction in a thin Insulated Ring.

$$\frac{\partial u}{\partial t} = k\,\frac{\partial^2 u}{\partial x^2}$$

$$u(-L,t) = u(L,t)$$

$$\frac{\partial u}{\partial x}(-L,t) = \frac{\partial u}{\partial x}(L,t)$$

$$u(x,0) = f(x).$$

$$\Rightarrow \quad \frac{d^2\phi}{dx^2} = -\lambda\phi$$

$$\phi(-L) = \phi(L)$$

$$\frac{\partial \phi(-L)}{\partial x} = \frac{\partial \phi(L)}{\partial x}$$

periodic boundry conditions.

- $\lambda > 0$,

$$\phi = C_1 \cos(\sqrt{\lambda}\, x) + C_2 \sin\sqrt{\lambda}\, x$$

$$\phi(-L) = \phi(L)$$

$$\Rightarrow \quad C_1 \cos(\sqrt{\lambda}\, L) + C_2 \sin(\sqrt{\lambda}\, L) = C_1 \cos(\sqrt{\lambda}\, L) - C_2 \sin(\sqrt{\lambda}\, L)$$

$$\Rightarrow \quad \boxed{2\, C_2 \sin(\sqrt{\lambda}\, L) = 0}$$

$$\frac{d\phi}{dx} = \sqrt{\lambda}\left(-C_1 \sin(\sqrt{\lambda}x) + C_2 \cos(\sqrt{\lambda}x)\right)$$

$$\Rightarrow \boxed{C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}L) = 0}$$

if $\sin(\sqrt{\lambda}L) \neq 0 \Rightarrow C_1 = C_2 = 0$

Thus, to find a nontrivial sol.

$$\sin(\sqrt{\lambda}L) = 0$$

$$\Rightarrow \lambda = \frac{n^2\pi^2}{L^2} \qquad n = 1, 2, \ldots$$

but here we have a very interesting case

both $\cos(\frac{n\pi x}{L})$ and $\sin(\frac{n\pi x}{L})$

are eigenfunctions of the same eigenvalue.

and any linear combination

$\lambda > 0$,

$$u(x,t) = \cos\frac{n\pi x}{L}\, e^{-(n\pi/L)^2 kt}$$

$$u(x,t) = \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

$\lambda = 0 \qquad \phi = C_1 + C_2 x$

$$u(x,t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt} + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

---

problem:

$$k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}$$

$0 < x < L \quad , t > 0$

I.C. $\quad u(x,0) = f(x)$

B.C $\quad \frac{\partial u(0,t)}{\partial x} = \frac{\partial u(L,t)}{\partial x} = 0$

Sol: $\quad u(x,t) = \phi(x)\, G(t)$

$$k\, G(t) \frac{\partial^2 \phi(x)}{\partial x^2} - h\, \phi(x) G(t) = \phi(x) \frac{\partial G(t)}{\partial t}$$

$$\frac{1}{\phi(x)}\frac{\partial^2\phi(x)}{\partial x^2} - \frac{h}{K} = \frac{1}{KG(t)}\frac{\partial G(t)}{\partial t}$$

$$\Rightarrow \quad \underbrace{\frac{1}{\phi(x)}\frac{\partial^2\phi(x)}{\partial x^2}} = \frac{h}{K} + \frac{1}{KG(t)}\frac{\partial G(t)}{\partial t} = -\lambda$$

$$\Rightarrow \quad \frac{\partial G(t)}{\partial t} = -(h + \lambda K)\, G(t)$$

$$\Rightarrow G(t) = e^{-(h+\lambda K)t}$$